

1. Convert each denary number into 8-bit binary. Show your working. [4]

(a) 91 = (b) 172 =

2. Give the denary value of each binary number. [2]

(a) 10110011 = (b) 01111110 =

3. State the number of unique binary values that can be represented in: [3]

(a) 8 bits (b) 16 bits (c) 24 bits

4. State the largest denary integer that can be represented as an unsigned integer in 16 bits. [1]

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5. Add the two 8-bit binary numbers 10011010 and 01100110. State the value the 8-bit register will hold after the addition, and explain why. [3]

6. A player's score is stored in an 8-bit register. The score is currently 250 and the player then earns 10 more points. State the value the register will now hold. Explain your answer. [2]



The ladder this week — Karel, not Python. No ice-cream challenge this lesson by design — this week's keyboard work is **Karel 05 LightTrail** and **06 TheLongHall**, your first loops. Binary is the paper half.

☛ **Want to dig?** Your questions 5 and 6 both walked a register off the end of its range. Now find **your own machine's edge**: open a phone calculator and keep squaring until it gives up, or type $=2^{1024}$ into a spreadsheet and then $=2^{1025}$. Note *exactly* where it breaks and what it says. Every device you own has this edge; the only difference between them is how far out it sits and how politely they tell you.

draft / working